

# Square Values

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## MA.8.AR.2.3

*I can square a number, find the side of a square from its area, and explain the graph of  $y = x^2$ .*

*A new unit, and a new benchmark.*

**Today everything is a square you could draw.**

# Warm-Up

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*On your own. Four minutes.*

## Same and different

What is the same about these two, and what is different?

$$7^2$$

$$(-5)^2$$

# Notes

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## I. Squaring means multiplying a number by itself

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- A. square (a number) — to multiply the number by itself
- B. **5 squared =  $5^2$  =  $5 \times 5$  = 25.**
- C. **It is NOT  $5 \times 2$ . Squaring and doubling are different operations.**

# Notes

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## II. The perfect squares, to 225

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- A. perfect square — a number you get by squaring a whole number
- B. **1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225**
- C. Know these on sight. Every question in this unit is answered faster if you do.

# Notes

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## III. A square's area runs both ways

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- A. **Four equal sides, so the area is  $A = s \times s = s^2$ .**
- B. Side to area:  $s = 6$  gives  $A = 6^2 = 36$  square units.
- C. **Area to side:  $A = 36$  means  $s \times s = 36$ , so  $s = 6$  units.**
- D. Backwards is the harder direction. Ask: what number times itself gives this?

# Notes

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## IV. Squaring never gives a negative answer

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- A. **A positive times itself is positive.  $3 \times 3 = 9$ .**
- B. **A negative times itself is positive.  $(-3) \times (-3) = 9$ .**
- C. So opposite x-values give the SAME y-value, and y is never negative.
- D. **The graph of  $y = x^2$  never goes below the x-axis. At  $x = 0$  it touches it.**

# Example 1

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*Nothing on paper yet. Answer out loud.*

| Kind of number | Side s        | What will you do? |
|----------------|---------------|-------------------|
| whole number   | 12 units      |                   |
| fraction       | $\frac{2}{5}$ |                   |
| decimal        | 0.7 units     |                   |

*Same operation three times. Say what changes.*

# Example 1

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*Now the areas. Whiteboards.*

| Kind of number | Side $s$      | Area $s^2$ |
|----------------|---------------|------------|
| whole number   | 12 units      |            |
| fraction       | $\frac{2}{5}$ |            |
| decimal        | 0.7 units     |            |

# Example 1

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*The areas, checked.*

| Kind of number | Side $s$      | Area $s^2$     |
|----------------|---------------|----------------|
| whole number   | 12 units      | 144            |
| fraction       | $\frac{2}{5}$ | $\frac{4}{25}$ |
| decimal        | 0.7 units     | 0.49           |

# Example 1

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*Now backwards.*

**A square has an area of 2.25 square units.**

**What is the length of one side?**

*No guessing. Show me how you narrowed it down.*

# Example 1

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*Narrow it down.*

| Try | Try × Try        | Compared with 2.25 |
|-----|------------------|--------------------|
| 1   | $1 \times 1 = 1$ | too small          |
| 2   | $2 \times 2 = 4$ | too big            |
| ?   | $? \times ?$     | 2.25 ✓             |

# Example 1

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*The side.*

$$1.5 \times 1.5 = 2.25$$

The side is 1.5 units.

# Example 2

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*Fill it in on your whiteboard. Seven values.*

| <b>x</b> | <b>y = x<sup>2</sup></b> |
|----------|--------------------------|
| -3       |                          |
| -2       |                          |
| -1       |                          |
| 0        |                          |
| 1        |                          |
| 2        |                          |
| 3        |                          |

# Example 2

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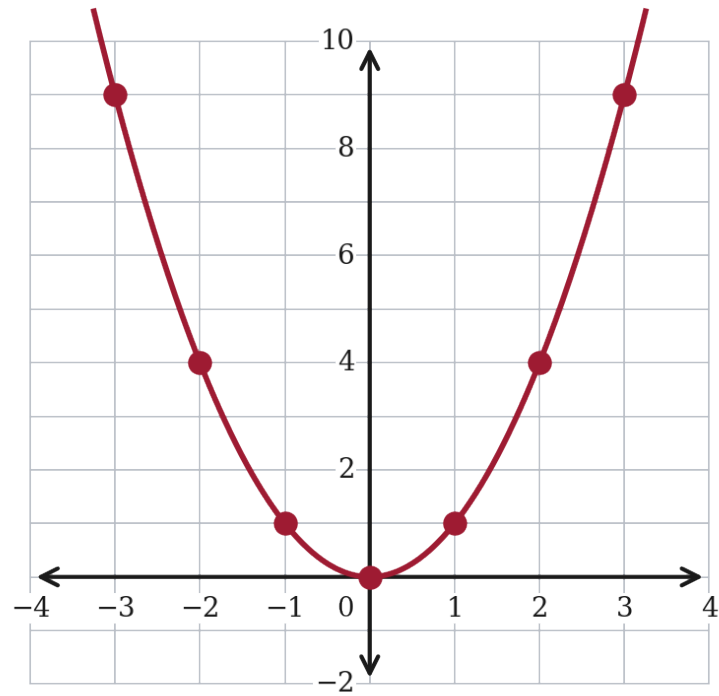
*The values.*

| <b>x</b> | <b><math>y = x^2</math></b> |
|----------|-----------------------------|
| -3       | 9                           |
| -2       | 4                           |
| -1       | 1                           |
| 0        | 0                           |
| 1        | 1                           |
| 2        | 4                           |
| 3        | 9                           |

# Example 2

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*Now the picture.*



# Example 3

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*The best question in this lesson.*

**Can a square have an area of 8 square units?**

*Yes or no, and how do you know?*

# Example 3

*Every whole number, tried.*

| Side | Side × Side  | Area          |
|------|--------------|---------------|
| 2    | $2 \times 2$ | 4   too small |
| 3    | $3 \times 3$ | 9   too big   |
| ?    | $? \times ?$ | 8   ✓         |

**So the side is between 2 and 3 — and it is not a whole number.**

# Example 3

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*And it is not a fraction either.*

**The side is about 2.83 — and that decimal never stops and never repeats.**

# Whiteboards • Question 1 of 9

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*Boards up on three.*

$$6^2$$

**Write the value.**

*One number.*

# Whiteboards • Question 1 of 9

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*Answer.*

$$6^2$$

$$6 \times 6$$

**Answer: 36**

# Whiteboards • Question 2 of 9

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*Boards up on three.*

$$(-6)^2$$

**Write the value.**

*One number.*

# Whiteboards • Question 2 of 9

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*Answer.*

$$(-6)^2$$

$$(-6) \times (-6)$$

**Answer: 36**

# Whiteboards • Question 3 of 9

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*Boards up on three.*

$$-6^2$$

**Write the value.**

*One number.*

# Whiteboards · Question 3 of 9

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*Answer.*

$$-6^2$$

$$-(6 \times 6)$$

**Answer: -36**

# Whiteboards • Question 4 of 9

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*Boards up on three.*

$$(0.6)^2$$

**Write the value.**

*One number.*

# Whiteboards · Question 4 of 9

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*Answer.*

$$(0.6)^2$$

$$0.6 \times 0.6$$

**Answer: 0.36**

# Whiteboards • Question 5 of 9

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*Boards up on three.*

$$\left(\frac{2}{3}\right)^2$$

**Write the value.**

*One number.*

# Whiteboards • Question 5 of 9

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*Answer.*

$$\left(\frac{2}{3}\right)^2$$

square the top AND the bottom

**Answer: 4/9**

# Whiteboards • Question 6 of 9

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*Boards up on three.*

$$s^2 = \frac{4}{9}$$

**Write the side s.**

*A square has area  $4/9$  square units, and a side is a length.*

# Whiteboards • Question 6 of 9

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*Answer.*

$$s^2 = \frac{4}{9}$$

what times itself gives 4/9 ?

**Answer:  $s = 2/3$**

# Whiteboards • Question 7 of 9

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*Boards up on three.*

Which is bigger:  $(-9)^2$  or  $-9^2$  ?

**Write both values, then say which is bigger.**

*Two values and a comparison.*

# Whiteboards • Question 7 of 9

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*Answer.*

Which is bigger:  $(-9)^2$  or  $-9^2$  ?

$$(-9)^2 = 81 \qquad -9^2 = -81$$

**Answer:  $(-9)^2$  is bigger**

# Whiteboards • Question 8 of 9

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*Boards up on three.*

1

8

16

20

45

64

**Which of these are perfect squares? Write every one that is.**

*There is more than one.*

# Whiteboards • Question 8 of 9

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*Answer.*

1

8

16

20

45

64

$$1 = 1 \times 1$$

$$16 = 4 \times 4$$

$$64 = 8 \times 8$$

**Answer: 1, 16 and 64**

# Whiteboards • Question 9 of 9

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*Take your time. Boards up when you have written it.*

Can a square have an area of 8 square units? If it can, name the two whole numbers its side length falls between. Explain how you know.

**Write your answer in sentences.**

*This one is written work. Show the two products you used.*

# Whiteboards • Question 9 of 9

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*Answer.*

Can a square have an area of 8 square units? If it can, name the two whole numbers its side length falls between. Explain how you know.

$2 \times 2 = 4$  is too small;  $3 \times 3 = 9$  is too big

**Answer: Yes — the side is between 2 and 3**

# IXL

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*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.